

🌟 Introduction to JavaScript 🌟

JavaScript



1 What is JavaScript?

JavaScript is a programming language that makes websites interactive and fun!

👉 Example: When you click a button and something happens — that is JavaScript working!

Think of JavaScript like electricity in a house. HTML builds the house, CSS paints it, and JavaScript makes things work!

JavaScript makes websites **interactive and dynamic**. It allows buttons to change content instantly, checks form inputs like email or password, shows correct or wrong answers in quizzes, and updates page content (like cart totals) automatically without refreshing the page.

2 Very Brief History of JavaScript

JavaScript was created in **1995** by Brendan Eich and was first called **Mocha**. It was later renamed JavaScript to make websites interactive.

In **1996**, it became standardized as **ECMAScript**, so all browsers could support it properly.

In the **2000s**, JavaScript started adding animations, dropdown menus, and form validations, making websites more interactive.

In the **2010s**, powerful libraries and frameworks like React, Angular, and Node.js made JavaScript suitable for building complex web apps.

3 Why Should We Learn JavaScript?

✓ To create games 🎮

✓ To build websites 🌐

✓ To make apps 📱

✓ It is fun and creative!

4 Basic Syntax (Rules of Writing Code)

JavaScript code is written inside `<script>` tags.

Every statement usually ends with a semicolon (;)

Example:

```
alert("Hello World!");
```

5 Variables (Storing Information)

What is a Variable?

A **variable** is like a **box** that stores information.

We can use or change this information later in the program.

👉 Example:

If we write:

```
let name = "Meera";
```

It means we stored the name **Meera** inside a box called **name**.

✂ **Ways to Create Variables**

There are **3 ways** to create variables:

- **var** → Old way (not used much now)
- **let** → Modern way (value can be changed later)
- **const** → Modern way (value cannot be changed)

👉 Example:

```
const pi = 3.14;
```

Here, the value of **pi** cannot be changed.

6 JavaScript Data Types

Data types tell us what kind of data we are storing.

🔗 Primitive Types (Simple Data)

These store **simple values**:

- **String** → Text (Example: "Hello")
- **Number** → Numbers (10, 25, 3.14)
- **Boolean** → Only true or false
- **Undefined** → A variable is created but no value is given
- **Null** → Empty on purpose
- **Symbol** → Special unique value (not needed for beginners)

Reference Types (Complex Data)

These store **more detailed information**:

- **Array** → Stores many values in one variable
Example: ["Apple", "Mango", "Banana"]
- **Object** → Stores information in key-value form
Example: {name: "Ayaan", age: 12}

Functions (Reusable Code)

A function is like a machine  that does a task.

A **function** is a block of code that does a task when we call it.

Think of it like a **machine** — when you press the button, it works!

Example:

```
function greet(user) {
  alert("Hello " + user + "!");
}
```

```
greet("Ayaan");
```

This will show: **Hello Ayaan!**

Types of Functions (Basic Idea)

1. **Simple Function** → Does a simple task
2. **Function with Parameters** → Takes input (like a name)
3. **Function with Return Value** → Sends back a result
4. **Arrow Function** → Short way of writing a function

5. **Function Expression** → Function stored in a variable
6. **Default Parameters** → Gives default value if no input is given
7. **Function Calling Another Function** → One function uses another

8 Variable Scope

Scope means where we can use a variable.

- Global Scope – Used anywhere.
- Local Scope – Used only inside a function .
 - Block Scope (Let and Const)-Accessible only inside {} Blocs

9 Events

What Are Events?

An **event** is something that happens on a website.

👉 Example:

- Clicking a button
- Pressing a key
- Moving the mouse
- Page loading

Events help the website **respond to user actions**.

⚙️ How Events Work

- 1** A user does something (like clicking a button).
- 2** The browser detects the action.
- 3** JavaScript runs a function (called an **event handler**).
- 4** The webpage changes or reacts.

This makes websites **interactive and dynamic**.

📁 Example: Click Event

- The **onclick** event listens for a button click.
- When the button is clicked, a function runs.
- The function can change text, show a message, or do something else.

Example idea:

Click a button → Message appears: "Button was clicked!"

Example 1: Click Event

Explanation

- The onclick attribute listens for a button click.
- When clicked, it runs the show Message() function.
- The function changes the paragraph text.

Example 2: Mouseover Event

Explanation

- When you move your mouse over the paragraph, the text color changes to green.

Example 3: Key Press Event

Explanation

- Every time the user types a letter and releases the key, the text below updates instantly.

Example 4: Using addEventListener() (Modern Way)

Explanation

- Instead of using HTML attributes like onclick, you can attach events directly in JavaScript using addEventListener().
- This method is cleaner and preferred for modern web development.

10 DOM Manipulation

DOM means Document Object Model.

It allows JavaScript to change HTML content.

Example: Changing text when button is clicked.

The DOM Tree

When a web page loads, the browser creates a tree-like representation of the HTML document.

Each part of the document are nodes in the tree:

Node	Description
Document	Owner of all nodes in the document
<html>	Element Node
<head>	Element Node
<body>	Element Node
<a>	Element Node
href	Attribute Node
<h1>	Element Node
My Header	Text Node

Accessing HTML Elements

The HTML DOM can be used to access HTML elements.

The most common way to access an HTML element is to use the id of the element:

```
<!DOCTYPE html>
<html>
<body>
<h1>JavaScript HTML DOM</h1>
<h2>The getElementById() Method</h2>

<p id="demo"></p>

<script>
// Access a paragraph Element
const myPara = document.getElementById("demo");

// Change the content of the Element
myPara.innerHTML = "Hello World!";
</script>

</body>
</html>
```

JavaScript HTML DOM

The getElementById() Method

Hello World!

In the example above, the getElementById method used id="demo" to find the element.

- id="demo" is an **HTML property**

- `getElementById()` is a **DOM Method**
- `innerHTML` is a **DOM Property**

1 1 Best Practices in JavaScript

- ✓ Use meaningful variable names.
- ✓ Write clean and simple code.
- ✓ Use comments to explain code.
- ✓ Comment your code
- ✓ Validate User Input
- ✓ Test Across Browsers

Hands on Activities

1. Button Click – Change Text

Goal: Understand the click event.

Activity:

- Create a button.
- When the user clicks the button, change the text on the page.

Code Example

```
<!DOCTYPE html>
<html>
<body>

<p id="message">Click the button to change this text.</p>
<button onclick="changeText()">Click Me</button>

<script>
function changeText() {
  document.getElementById("message").innerHTML = "You clicked the button!";
}
</script>
```

```
</body>
</html>
```

What you learn

- Event handling
- onclick
- DOM manipulation

2. Mouse Hover Color Change

Goal: Learn mouseover event.

Activity:

- When the mouse moves over the text, change its color.

Code Example

```
<p id="text" onmouseover="changeColor()">Hover over me</p>
```

```
<script>
function changeColor() {
  document.getElementById("text").style.color = "green";
}
</script>
```

Challenge:

Make the color return to normal when the mouse leaves.

3. Live Typing Preview

Goal: Practice keyup event.

Activity:

- As the user types in an input box, display the typed text below.

Code Example

```
<input type="text" id="inputBox" onkeyup="showText()" placeholder="Type here">
<p id="output"></p>
```

```
<script>
function showText() {
  let text = document.getElementById("inputBox").value;
  document.getElementById("output").innerText = "You typed: " + text;
}
```



```
}  
</script>
```

What you learn

- Real-time updates
- Input events

4. Modern Event Handling (Best Practice)

Goal: Learn `addEventListener()`.

Activity:

- Handle button clicks using JavaScript instead of HTML attributes.

Code Example

```
<button id="btn">Click Me</button>  
<p id="result"></p>
```

```
<script>  
document.getElementById("btn").addEventListener("click", function() {  
  document.getElementById("result").innerText = "Event handled using addEventListener!";  
});  
</script>
```

What you learn

- Modern JavaScript event handling

5. Mini Activity: Simple Counter App

Goal: Combine events + logic.

Activity:

- Create a button that increases a number each time you click.

Code Example

```
<h2 id="count">0</h2>  
<button onclick="increase()">Increase</button>
```

```
<script>  
let counter = 0;  
  
function increase() {
```

```
counter++;  
document.getElementById("count").innerText = counter;  
}  
</script>
```

Extra Challenge

- Add a **Decrease button**
- Add a **Reset button**

6. Fun Beginner Project (Very Useful)

Color Changer

Click a button to change the page background.

```
<button onclick="changeBg()">Change Background</button>
```

```
<script>  
function changeBg() {  
  document.body.style.backgroundColor = "lightblue";  
}  
</script>
```

Challenge:

Make it change to **random colors**.